WASP-11b

R-1650

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-10_160924` · created 2026-08-02T08:07:42+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457655.778 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.05
Transit depth (Rp/Rs)^2	3.04 +/- 1.75 [%]
Orbital Inclination (inc)	89.5 +/- 1.72
Airmass coefficient 1 (a1)	0.89 +/- 0.82
Airmass coefficient 2 (a2)	0.07 +/- 0.15
Scatter in the residuals of the lightcurve fit is	129.67 %
Best Comparison Star	#2 - [105, 456]
Optimal Aperture	0.87
Optimal Annulus	3.75
Transit Duration (day)	0.105 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.05 vs 0.131 (deviation 0.69σ)
Duration fit vs published	0.105 d vs 0.1075 d (deviation 0.17σ)
Mid-time O–C	2.6 min
Depth SNR	2.8
Residual scatter	129.67 %

Light curve

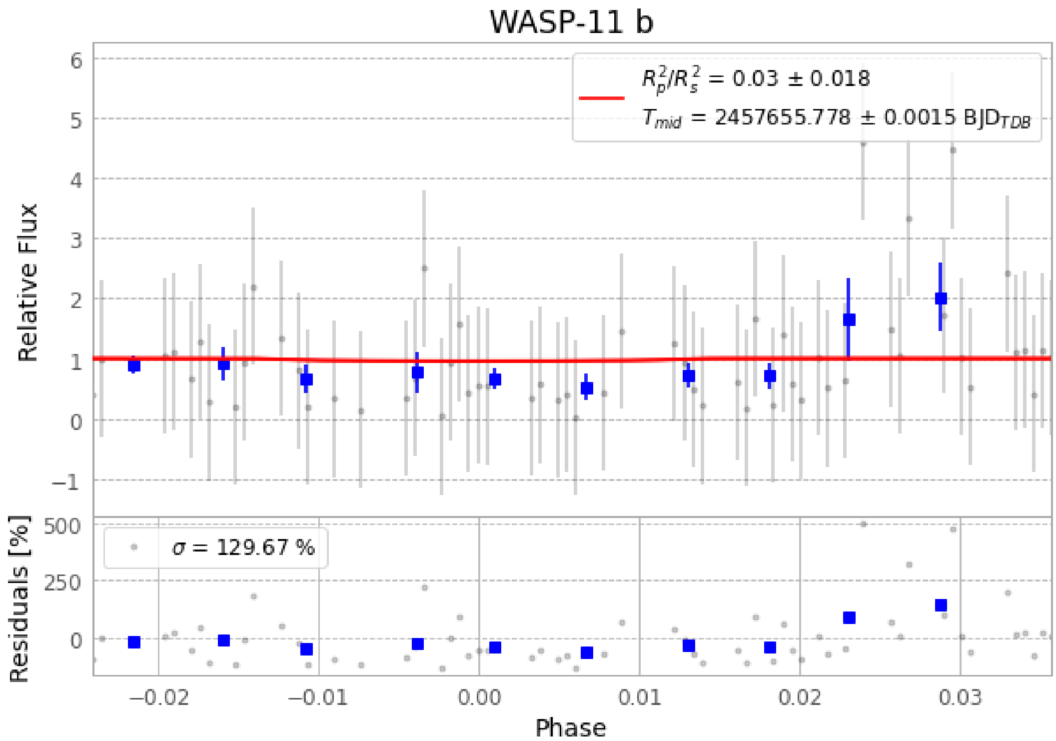

Figure 1 — FinalLightCurve_WASP-11 b_2016-09-23.png (SHA-256 741c49265533ede7…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 439], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 009a533c09f438a2…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

109 FITS frames (72 MB), `HATP-10160924042112.FITS` → `HATP-10160924094512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-160924.log (cc15320f94bb…), FinalLightCurve_WASP-11 b_2016-09-23.png (741c49265533…), FinalLightCurve_WASP-11 b_2016-09-23.csv (42bf078ea220…)

Compute resources

Wall time 260.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)